

COMP9444: Neural Networks and Deep Learning

Project Description List

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001. Style Transfer: Transforming Horses into Zebras

Area of Research: Generative AI

Problem Statement: Style transfer is a fascinating technique that allows for the transformation of an image's style while preserving its essential content. Rapid and accurate application of style transfer is essential for various creative and practical applications, such as generating art, enhancing images, and transforming animal appearances. One intriguing application of this technique is converting images of horses into images of zebras. However, achieving realistic transformations while maintaining the structural integrity of the original image can be challenging. The goal of this project is to explore and implement an advanced style transfer algorithm that can accurately transform horse images into zebra images, providing high-quality, realistic results.

Dataset:

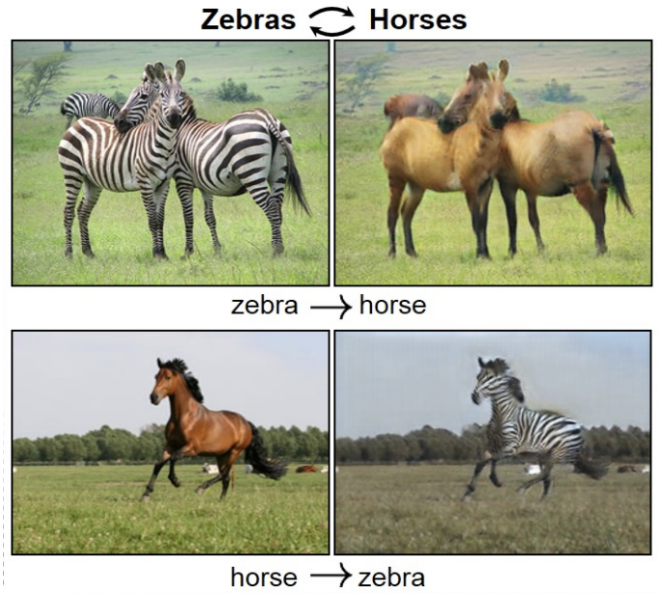


Figure 1: Example images if the zebra2horse and horse2zebra task, with dataset sample for zebra and horse.

Horse2zebra dataset [1] is a benchmark dataset for style transfer. The details about the dataset can be found in [1] and be downloaded from the URL given below:
Dataset URL: https://www.tensorflow.org/datasets/catalog/cycle_gan

Task: To develop an automatic horse-to-zebra transformation system using neural networks and deep learning that provides high accuracy, maintains the structural integrity of the original image, and produces realistic transformations. The system should be robust to varying appearances of horses and zebras and efficient enough to look as real as possible.

Relevant Papers

[1]. Zhu, J. Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired image-to-image translation using cycle-consistent adversarial networks. In Proceedings of the IEEE international conference on computer vision (pp. 2223-2232). https://openaccess.thecvf.com/content_iccv_2017/html/Zhu_Unpaired_Image-To-Image_Translation_ICCV_2017_paper.html
[2]. Tang, H., Liu, H., Xu, D., Torr, P. H., & Sebe, N. (2021). Attentiongan: Unpaired image-to-image translation using attention-guided generative adversarial networks. IEEE transactions on neural networks and learning systems, 34(4), 1972-1987. https://ieeexplore.ieee.org/abstract/document/9527389
[3]. Kim, T., Cha, M., Kim, H., Lee, J. K., & Kim, J. (2017, July). Learning to discover cross-domain relations with generative adversarial networks. In *International conference on machine learning* (pp. 1857-1865). PMLR.
https://proceedings.mlr.press/v70/kim17a

002. Environmental Sound Classification

Area of Research: Computer Vision (Audio Signal Processing)

Problem Statement: Environmental sound classification is a challenging task in the field of audio signal processing and machine learning. It has various applications, such as acoustic monitoring, surveillance systems, and smart home assistants. However, accurately classifying environmental sounds can be difficult due to the wide variety of sound events, background noise, and acoustic variations. This project aims to address this challenge by leveraging deep learning techniques, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to develop a robust and efficient system for classifying environmental sounds based on audio features. By training a deep learning model on the ESC-50 dataset, the goal is to create a system that can accurately identify different types of environmental sounds, enabling better understanding and interpretation of acoustic scenes.

Dataset: The dataset consists of 5-second-long recordings organized into 50 semantical classes (with 40 examples per class) loosely arranged into 5 major categories

Animals	Natural soundscapes & water sounds	Human, non-speech sounds	Interior/domestic sounds	Exterior/urban noises
Dog	Rain	Crying baby	Door knock	Helicopter
Rooster	Sea waves	Sneezing	Mouse click	Chainsaw
Pig	Crackling fire	Clapping	Keyboard typing	Siren
Cow	Crickets	Breathing	Door, wood creaks	Car horn
Frog	Chirping birds	Coughing	Can opening	Engine
Cat	Water drops	Footsteps	Washing machine	Train
Hen	Wind	Laughing	Vacuum cleaner	Church bells
Insects (flying)	Pouring water	Brushing teeth	Clock alarm	Airplane
Sheep	Toilet flush	Snoring	Clock tick	Fireworks
Crow	Thunderstorm	Drinking, sipping	Glass breaking	Hand saw

Figure 2: Dataset categories.

Dataset URL: https://github.com/karolpiczak/ESC-50

Task: Develop a deep learning-based environmental sound classification model using CNNs and RNNs to accurately classify audio clips into their respective sound categories.

Relevant Papers:

[1]. Piczak, K. J. (2015). ESC: Dataset for environmental sound classification. In Proceedings of the 23rd ACM international conference on Multimedia (pp. 1015-1018). http://karol.piczak.com/papers/Piczak2015-ESC-Dataset.pdf

[2]. Salamon, J., & Bello, J. P. (2017). Deep convolutional neural networks and data augmentation for environmental sound classification. IEEE Signal Processing Letters, 24(3), 279-283. https://doi.org/10.1109/LSP.2017.2657381

[3]. Zhang, Z., Xu, S., Zhang, S., Qiao, T., & Cao, S. (2019). Learning attentive representations for environmental sound classification. IEEE Access, 7, 130327-130339. https://doi.org/10.1109/ACCESS.2019.2939495

[4]. Boddapati, V., Petef, A., Rasmusson, J., & Lundberg, L. (2017). Classifying environmental sounds using image recognition networks. Procedia Computer Science, 112, 2048-2056. https://doi.org/10.1016/j.procs.2017.08.250

003. Question Answering on Movie Knowledge Graphs

Area of Research: Natural Language Processing (NLP).

Problem Statement: Question Answering is the task of automatically generating correct answers to factual questions. The task has numerous practical applications, including the development of Recommender Systems, Digital Assistants and Customer Service systems. This project deals with Question Answering in the Movie Domain: the objective is to train a model on a large Knowledge Graph of movie facts, so that it learns to predict the answer to unseen questions in the test set.
Dataset:

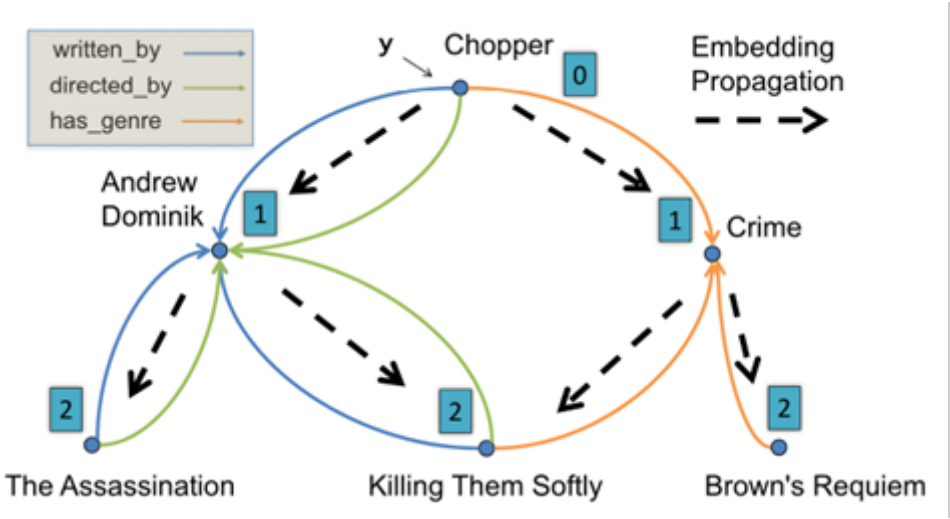


Figure 3: **MetaQA** (MoviE Text Audio QA) is a dataset of over 400,000 question-answer pairs on the movie domain. Questions in the corpus include identifying the cast of the movie, the director, the genre or the year it was released in.

Dataset URL: https://github.com/yuyuz/MetaQA

Dataset Paper: https://arxiv.org/pdf/1709.04071

Task: Train a model to extract relevant knowledge from graph-structured movie facts, so that it learns to generate the correct response to factual questions about movies.

Relevant Papers

[1].Gupta, Aayushee, et al. "An Inference Approach To Question Answering Over Knowledge Graphs." arXiv preprint arXiv:2112.11070 (2021). https://arxiv.org/pdf/1709.04071

[2]. Saxena, Apoorv, Aditay Tripathi, and Partha Talukdar. "Improving multi-hop question answering over knowledge graphs using knowledge base embeddings." Proceedings of the 58th annual meeting of the association for computational linguistics. 2020. https://aclanthology.org/2020.acl-main.412.pdf

[3]. Sun, Haitian, Tania Bedrax-Weiss, and William Cohen. "PullNet: Open Domain Question Answering with Iterative Retrieval on Knowledge Bases and Text." Proceedings of EMNLP-IJCNLP. 2019.. https://arxiv.org/pdf/1904.09537

004. GAN-Based Semantic Segmentation Enhancement for Complex Scenes in ADE20K

Area of Research: Computer Vision (CV)

Problem Statement:
This project aims to enhance the accuracy and detail of semantic segmentation for complex scenes in the ADE20K dataset using Generative Adversarial Networks (GANs). By leveraging the capabilities of GANs, the project will focus on refining segmentation maps to capture fine-grained details and improve overall segmentation performance. The approach involves training a GAN-based model that learns to generate high-quality segmentation masks from input images, effectively addressing challenges such as occlusions, varying lighting conditions, and diverse scene compositions. The enhanced segmentation maps produced by the GAN will be evaluated against standard benchmarks to demonstrate improvements over traditional segmentation methods.
Dataset:

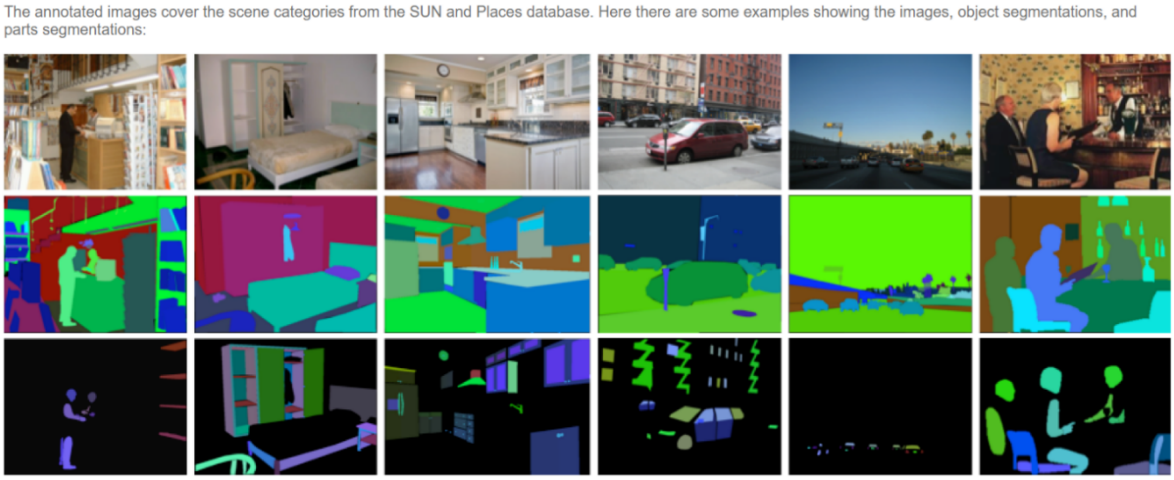


Figure 4: The annotated images cover the scene categories from the SUN and Places database. Here there are some examples showing the images, object segmentations, and parts segmentations.

ADE20K is a dataset for instance segmentation, semantic segmentation, and object detection tasks. It is applicable or relevant across various domains. Also, it is used in the robotics industry.

Dataset URL: https://datasetninja.com/ade20k

https://groups.csail.mit.edu/vision/datasets/ADE20K/

Task: The objective of this project is to develop a GAN-based model to enhance semantic segmentation accuracy for complex scenes in the ADE20K dataset. Traditional segmentation methods often struggle with intricate details and diverse scene elements, leading to suboptimal results. This project will utilize the generative capabilities of GANs to refine and improve segmentation maps, focusing on capturing fine-grained details and handling challenging conditions such as occlusions and varying lighting. By training the GAN on ADE20K, the model will learn to produce high-quality segmentation masks from input images. The effectiveness of the enhanced segmentation will be evaluated through standard metrics, demonstrating its superiority over conventional methods.

Relevant papers:
[1]. Abdollahi, Abolfazl, et al. "Improving road semantic segmentation using generative adversarial network." IEEE Access 9 (2021): 64381-64392.
[2]. Zhang, Chaoyi, et al. "MS-GAN: GAN-based semantic segmentation of multiple sclerosis lesions in brain magnetic resonance imaging." 2018 Digital Image Computing: Techniques and Applications (DICTA). IEEE, 2018.
[3]. Pan, Zhihao, et al. "Automatic pavement crack segmentation using a generative adversarial network (GAN)-based convolutional neural network." Results in Engineering 19 (2023): 101267.

005. Pedestrian Trajectory Prediction in Crowds

Area of Research: Trajectory Prediction, Sequence Modelling, Generative Model

Problem Statement: Pedestrian behaviour modelling and analysis is important for crowd scene understanding and has various applications in video surveillance, autonomous vehicles and social robotics. For example, pedestrians follow different trajectories to avoid collisions with neighbours and static obstacles. The goal of this project is to build a pedestrian trajectory prediction system using the GC Dataset with the requirements: **given a 3.2 second of historical trajectory data (2D coordinate sequences) and video frames, the system predicts the future path for the next 4.8 seconds.** To make accurate predictions, there are some advanced directions, including (1) the social interaction modelling among pedestrians, e.g., group detection; (2) the scene interaction modelling between pedestrians and static environment; (3) multi-future trajectory prediction using generative models (e.g., GAN).

Dataset:



Figure 5: Trajectory Prediction (left), Grand Central Station (GC) dataset (mid) and example of predictions (right).

Dataset URL: https://www.dropbox.com/s/7y90xsxq0l0yv8d/cvpr2015_pedestrianWalkingPathDataset.rar?e=1&dl=0
Dataset Paper: https://ieeexplore.ieee.org/document/7298971/

Task: This task is to build a pedestrian trajectory prediction system using the GC Dataset with the basic requirements: **given a 3.2 second of historical trajectory data and video frames, the system makes accurate and socially acceptable predictions for the 4.8 seconds.** Commonly used evaluation metrics are:

- Average Displacement Error: The average L2 distance between the predictions and the ground truth.
- Collision Rate (optional): whether the collisions exist (e.g., distance < 0.2m) among predictions.

Relevant Papers

[1]. B. Zhou, X. Wang and X. Tang. "Understanding Collective Crowd Behaviors: Learning a Mixture Model of Dynamic Pedestrian-Agents." in Proceedings of IEEE Conference on Computer Vision and Pattern Recognition, 2012.

[2]. S. Yi, H. Li and X. Wang, "Understanding pedestrian behaviors from stationary crowd groups," 2015 IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 3488-3496.

[3]. Y. Xu, Z. Piao and S. Gao, "Encoding Crowd Interaction with Deep Neural Network for Pedestrian Trajectory Prediction," 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 5275-5284.

[4] Gupta, Agrim, et al. "Social gan: Socially acceptable trajectories with generative adversarial networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.

006. Remote Sensing Image Caption Generation Using the RSICD Dataset

Area of Research: Computer Vision and Natural Language Processing

Problem Statement: Remote sensing images captured by satellites and airplanes contain complex scenes that are difficult to describe with simple labels. Accurate and flexible descriptions of these images are crucial for applications such as environmental monitoring, urban planning, and disaster management. The challenge lies in generating detailed and coherent sentences that capture the semantics of remote sensing images, considering scale, category, and rotation ambiguities. This project aims to develop a deep learning model for generating captions for remote sensing images using the RSICD dataset, advancing the task of remote sensing image captioning.

Dataset: RSICD (Remote Sensing Image Captioning Dataset) [1]

The RSICD dataset consists of more than 10,000 remote sensing images, each annotated with five descriptive sentences. The images are sourced from platforms like Google Earth, Baidu Map, MapABC, and Tianditu, and are fixed at 224 ? 224 pixels. The dataset includes a diverse range of scenes and objects, making it a comprehensive resource for developing and evaluating image captioning models.



1. An old court is surrounded by white houses.
2. A playground is surrounded by many trees and long buildings.
3. A playground with basketball fields next to it is surrounded by many green trees and buildings.
4. Many green trees and several long buildings are around a playground.
5. This narrow, oval football field and closing basketball court, tennis court, parking lot together form this area, with plants wreathing it.



1. Four planes are stopped on the open space between the parking lot.
2. Four white planes are between two white buildings.
3. Some cars and two buildings are near four planes.
4. Four planes are parked next to two buildings on an airport.
5. Four white planes are between two white buildings.

Figure 6: Example of images and corresponding five sentences each image selected from our data set.

Dataset URL: https://github.com/201528014227051/RSICD_optimal

Tasks: Develop an automated system for generating descriptive captions for remote sensing images using the RSICD dataset. The objectives are:

- To create a model that generates accurate and coherent captions for remote sensing images.
- To address the challenges of scale, category, and rotation ambiguities inherent in remote sensing images.

Relevant Papers

[1]. Lu, X., Wang, B., Zheng, X., & Li, X. (2018). "Exploring Models and Data for Remote Sensing Image Caption Generation." IEEE Transactions on Geoscience and Remote Sensing, 56(4), 2183-2195.

007. Images Style Transfer

Area of Research: Computer Vision (CV)

Problem Statement: Image style transfer applies the artistic style of a reference image to a target image while preserving its content. Traditional CNN-based methods struggle to capture global image dependencies due to their local receptive fields, leading to biased content representations and reduced transfer quality. Recent advances in Generative AI (GenAI) and GANs offer promising alternatives, provided the models can integrate both local details and global information, supported by scale-invariant positional encoding suited to image-based tasks. This project explores GenAI- and GAN-based approaches to improve content preservation and style quality, evaluated qualitatively and quantitatively against state-of-the-art techniques.

Task: Develop an automatic image style transfer method using GenAI or GANs that achieves high accuracy and robustness in transferring artistic styles while preserving original content, enhancing applicability in art, design, and media production.

Dataset:

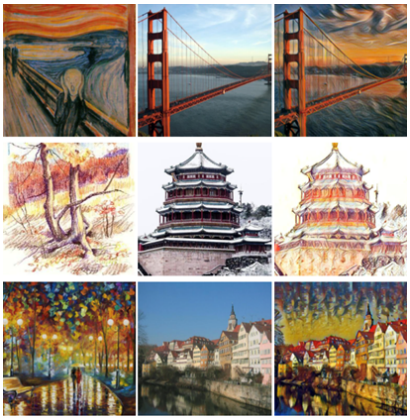


Figure 7: COCO examples: style images (first column), content images (second column), and style transfer results (last column).

The MS COCO dataset [1] is a large-scale benchmark (164,000 images) widely used for object detection, segmentation, key-point detection, and captioning.
Style Images: WikiArt | Content Dataset: COCO 2014 (Kaggle)

Relevant Papers

[1]. Deng, Y., et al. (2022). StyTr^2: Image style transfer with transformers. *CVPR* (pp. 11326–11336). [PDF]
[2]. Xu, W., et al. (2021). DRB-GAN: A dynamic ResBlock generative adversarial network for artistic style transfer. *ICCV* (pp. 6383–6392). [PDF]
[3]. Chen, X., et al. (2018). Gated-GAN: Adversarial gated networks for multi-collection style transfer. *IEEE TIP*, 28(2), 546–560. [Link]
[4]. Liu, H., et al. (2018). Artsy-GAN: A style transfer system with improved quality, diversity and performance. *ICPR* (pp. 79–84). [Link]
[5]. Xu, W., et al. (2023). Learning dynamic style kernels for artistic style transfer. *CVPR* (pp. 10083–10092). [PDF]

008. Realistic Face Generation

Area of Research: Computer Vision, Generative AI

Problem Statement: Generating realistic human face images is a crucial task in computer vision and generative AI, with applications in entertainment, digital avatars, privacy-focused AI, and synthetic data generation. This project aims to develop a deep generative model leveraging Generative Adversarial Networks (GANs) or Diffusion Models to produce diverse, high-quality synthetic faces. Its contributions extend to digital content creation, gaming, virtual reality, and marketing, reducing the reliance on real human photography. It also supports AI ethics by promoting privacy-preserving synthetic data, advances GAN and diffusion model research, and aids industries such as fashion and security in biometric authentication and deepfake detection.



Figure 8: The CelebA dataset consists of 202,599 celebrity face images with 40 annotated attributes (e.g., gender, smiling, glasses, hair color) and 5 key facial landmarks. It provides a diverse collection of face images under different poses, lighting conditions, and expressions.

Dataset URL: https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html

Task: Develop a Generative AI model for realistic face generation based on the CelebA dataset. The project may involve:

1. Implementing Generative Adversarial Networks (GANs) or Diffusion Models.
2. Experimenting the difference of GANs and Diffusion Models.
3. Leveraging conditional generative model to generate faces with specific attributes (e.g., smiling, age, glasses).

Relevant Papers

[1]. Liu, Z., Luo, P., Wang, X., & Tang, X. (2015). Deep learning face attributes in the wild. In *Proceedings of the IEEE international conference on computer vision* (pp. 3730-3738).

https://liuziwei7.github.io/projects/FaceAttributes.html

[2]. Karras, T. (2019). A Style-Based Generator Architecture for Generative Adversarial Networks. *arXiv preprint arXiv:1812.04948*.

https://arxiv.org/abs/1812.04948

[3]. Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in neural information processing systems*, *33*, 6840-6851.

https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html

009. Generative Adversarial Network (GAN) medical image segmentation

Area of Research: Computer vision

Problem Statement: The project aims to investigate the usefulness of GANs in medical image segmentation. With the rise of Diffusion Networks, GANs have largely fallen out of favour in generative AI tasks other than as baseline comparisons. Hence, the project's purpose is to examine the effectiveness of repurposing GANs for image segmentation on a Brain Tumour Segmentation dataset that was investigated using other image segmentation techniques.

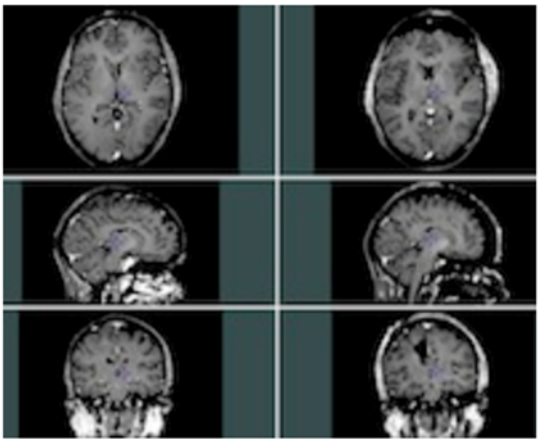


Figure 9: Dataset Example

Dataset URL: https://nist.mni.mcgill.ca/bite-brain-images-of-tumors-for-evaluation-database/

Task: Students will investigate the application of GANs to the relatively unknown BITE dataset. Before training the neural networks on the dataset, students will need to follow the **first** relevant paper closely in generating the dataset from dataset URL to emulate the experiment covered in the paper. Students should then review the **second** relevant paper and choose a relatively simple GAN to apply to the dataset before experimenting with more complex models.

Relevant Papers

[1]. Walsh, J., Othmani, A., Jain, M., & Dev, S. (2022). Using U-Net network for efficient brain tumor segmentation in MRI images. *Healthcare Analytics*, 2, 100098. https://doi.org/10.1016/j.health.2022.100098

[2]. Xun, S., Li, D., Zhu, H., Chen, M., Wang, J., Li, J., Chen, M., Wu, B., Zhang, H., Chai, X., Jiang, Z., Zhang, Y., & Huang, P. (2022). Generative adversarial networks in medical image segmentation: A review. *Computers in Biology and Medicine*, 140, 105063. https://doi.org/10.1016/j.compbiomed.2021.105063

010. Cross-modality Person Re-identification

Area of Research: Computer Vision

Problem Statement: Person re-identification (Re-ID) is an important problem in video surveillance because it enables tracking and identifying individuals across different camera views, enhancing security and public safety. It addresses challenges like varying lighting, occlusions, and changes in appearance, which are critical for crime prevention, suspect tracking, and efficient management of large surveillance networks. The goal of this project is to develop a neural network to accurately match and identify individuals across multiple camera views in different locations or scenarios.

Dataset:

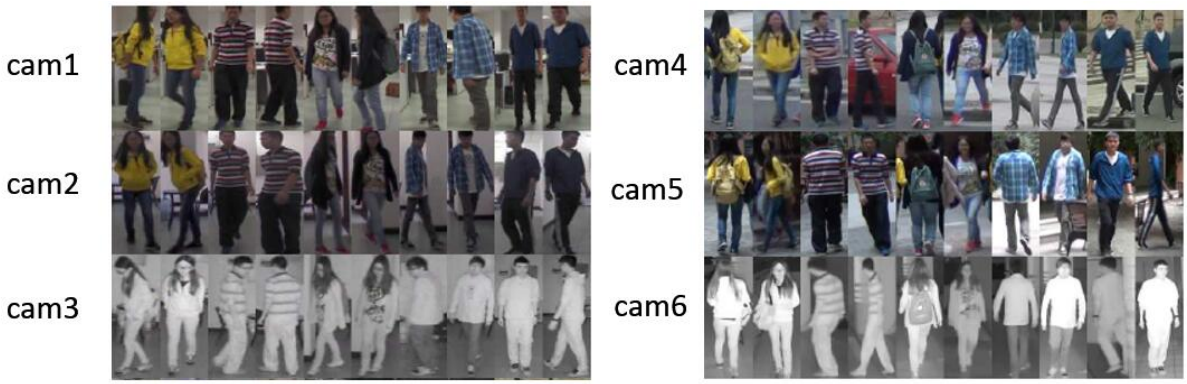


Figure 10: Examples for the 6 cameras in SYSU-MM01 dataset. It includes 491 identities with 30,071 RGB images and 15,792 IR images from 4 RGB and 2 IR cameras, designed for multimodal re-identification in varied lighting.

SYSU-MM01dataset [1] is an RGB-Infrared (IR) multi-modality pedestrian dataset for cross-modality person re-identification. The details about the dataset can be found in [1] and be downloaded from the URL given below:
Dataset URL:
https://www.kaggle.com/datasets/coconutjean/sysumm01/data

Task: Development of a neural network framework to extract suitable features for images of different modalities from the 6 cameras to be able to re-identify them successfully. This framework aims to make the features extracted from images of the same person as similar as possible, and the features of different people dissimilar.

Relevant Papers

[1]. Wu, A., Zheng, W. S., Gong, S., & Lai, J. (2020). RGB-IR person re-identification by cross-modality similarity preservation. *International journal of computer vision*, *128*(6), 1765-1785. Paper Link

[2]. Hao, X., Zhao, S., Ye, M., & Shen, J. (2021). Cross-modality person re-identification via modality confusion and center aggregation. In *Proceedings of the IEEE/CVF International conference on computer vision* (pp. 16403-16412). Paper Link

[3]. Fu, C., Hu, Y., Wu, X., Shi, H., Mei, T., & He, R. (2021). CM-NAS: Cross-modality neural architecture search for visible-infrared person re-identification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 11823-11832). Paper Link

[4]. Ye, M., Ruan, W., Du, B., & Shou, M. Z. (2021). Channel augmented joint learning for visible-infrared recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 13567-13576). Paper Link

[5]. Lu, H., Zou, X., & Zhang, P. (2023, June). Learning progressive modality-shared transformers for effective visible-infrared person re-identification. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 37, No. 2, pp. 1835-1843). Paper Link

011. Semantic Segmentation and Instance-Level Labeling in Urban

Area of Research : Computer Vision

Problem Statement: Understanding complex urban street scenes is critical for various applications. Object detection has seen significant progress thanks to large-scale datasets, particularly with the advent of deep learning. However, existing datasets fall short in capturing the full complexity of real-world urban environments for semantic scene understanding. To bridge this gap, we present *Cityscapes*, a comprehensive benchmark suite and large-scale dataset designed for training and evaluating pixel-level and instance-level semantic labelling methods.

Dataset:



Figure 11: Example images of the Cityscapes dataset with annotations.

Cityscapes is a large-scale database which focuses on semantic understanding of urban street scenes. It provides semantic, instance-wise, and dense pixel annotations for 30 classes grouped into 8 categories (flat surfaces, humans, vehicles, constructions, objects, nature, sky, and void). The dataset consists of around 5000 fine annotated images and 20000 coarse annotated ones.

URL - https://www.cityscapes-dataset.com/register/

Task: Develop and evaluate a deep learning model for pixel-level and instance-level semantic segmentation using the Cityscapes dataset.

Relevant Research Papers:

[1]. Liu, Chenxi, Liang-Chieh Chen, Florian Schroff, Hartwig Adam, Wei Hua, Alan L. Yuille, and Li Fei-Fei. "Auto-deeplab: Hierarchical neural architecture search for semantic image segmentation." In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 82-92. 2019.
https://arxiv.org/abs/1901.02985

[2]. Chen, Liang-Chieh, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L. Yuille. "Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs." *IEEE transactions on pattern analysis and machine intelligence* 40, no. 4 (2017): 834-848. https://arxiv.org/abs/1606.00915

012. Automated Radiology Report Generation using Vision-Language Models

Area of Research: Natural Language Processing (NLP) & Computer Vision

Problem Statement: Medical radiology reports are essential for diagnosing diseases but generating detailed and accurate reports requires expert radiologists. Due to the increasing demand for radiological assessments, there is a growing need for automated systems that can generate medical reports from radiology images. Recent advancements in Vision-Language Models (VLMs), such as Transformer-based architectures, have shown potential in automating medical report generation. However, challenges such as medical terminology accuracy, report consistency, and data bias must be addressed. This project aims to develop a deep learning-based system that generates automated radiology reports from chest X-ray images. The system will use a transformer-based vision-language model to improve the accuracy and coherence of the generated reports.

Dataset:

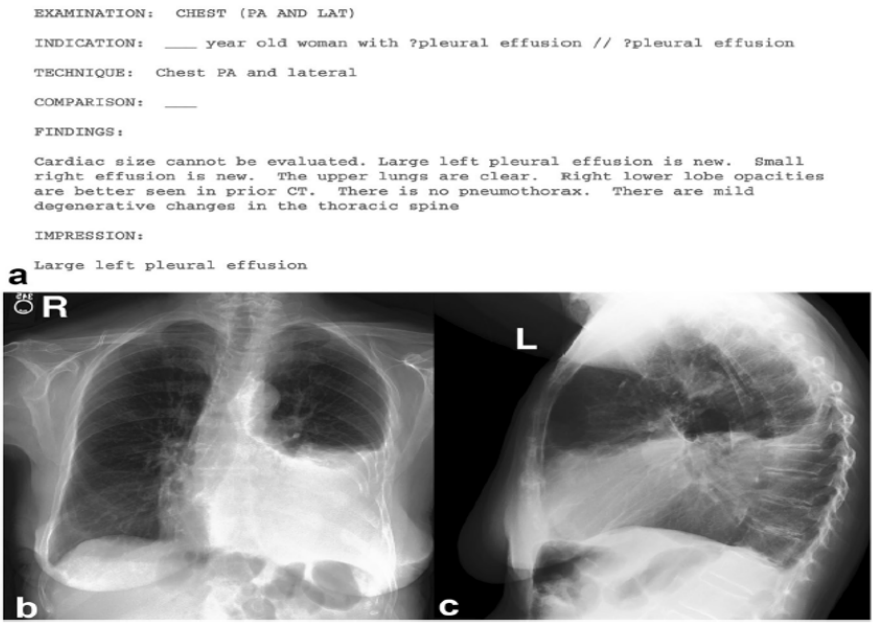


Figure 12: Example study contained in MIMIC-CXR. Above (a), the radiology report provides the interpretation of the image. PHI has been removed and replaced with three underscores (___). Below, the two chest radiographs for this study are shown: (b) the frontal view (left image) and (c) the lateral view (right image).

The MIMIC-CXR dataset is a publicly available dataset containing chest X-ray images and corresponding radiology reports. The dataset includes 377,110 images and 227,835 reports.
Dataset URL: https://www.kaggle.com/datasets/simhadrisadaram/mimic-cxr-dataset?resource=download&select=mimic_cxr_aug_validate.csv

Tasks: Develop a vision-language model to generate radiology reports from chest X-ray images. The objectives are:

- Fine-tune a transformer-based vision-language model on the MIMIC-CXR dataset.
- Improve the clinical relevance and coherence of generated reports.
- Evaluate model performance using BLEU, ROUGE, and clinical accuracy metrics.

Relevant Papers:

[1] Boag, W., McDermott, M., Saeed, M., et al. "MIMIC-CXR: A large publicly available database of labeled chest radiographs," Scientific Data, 2019.

013. Named Entity Recognition (NER) for Biomedical Texts using Large Language Models (LLMs)

Area of Research: Natural Language Processing (NLP)

Problem Statement: Biomedical Named Entity Recognition (NER) is a critical task in NLP, involving the identification of medical terms such as diseases, drugs, genes, and anatomical entities in unstructured clinical texts. Traditional rule-based and statistical approaches struggle to generalize across different biomedical domains. Recent transformer-based Large Language Models (LLMs) such as BioBERT and PubMedBERT have shown promising results in improving entity recognition accuracy. This project aims to develop and fine-tune a transformer-based NER model for biomedical text processing, focusing on extracting key medical concepts from clinical notes and research papers.

Dataset:

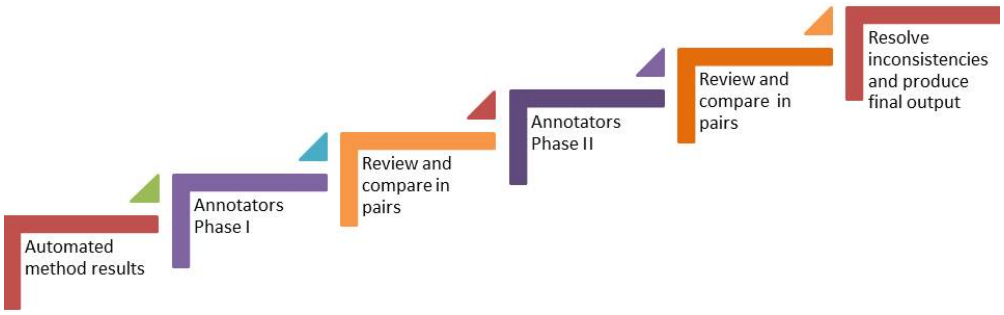


Figure 13: The illustration of the annotation process involving 12 annotators working on pairs on 793 PubMed abstracts for disease name recognition covering all the sentences in every PubMed citation.

NCBI Disease Corpus
The NCBI Disease Corpus contains 793 PubMed abstracts annotated for disease mentions and entity linking to the Medical Subject Headings (MeSH) terminology. It serves as a benchmark dataset for biomedical NER.

Dataset URL: https://www.ncbi.nlm.nih.gov/CBBresearch/Dogan/DISEASE/

Tasks: Develop a biomedical NER model using transformer-based architectures. The objectives are:

- Fine-tune BioBERT or PubMedBERT on biomedical text datasets.
- Improve generalization using domain adaptation techniques.
- Evaluate model performance using precision, recall, and F1-score.

Relevant Papers:

[1] Lee, J., Yoon, W., Kim, S., et al. "BioBERT: A pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, 2020.

014. Automated Marine Debris Detection using Deep Learning

Learning

Area of Research: Computer Vision, Deep Learning, Environmental

Problem Statement: Marine debris poses a significant threat to ocean ecosystems, harming marine life and disrupting the natural balance. Traditional cleanup efforts rely on human intervention, which is costly and inefficient. With advancements in AI, automated systems can now detect and classify underwater debris, assisting in large-scale marine conservation efforts. The goal is to develop a deep learning-based system to accurately identify and localize marine debris using image data.

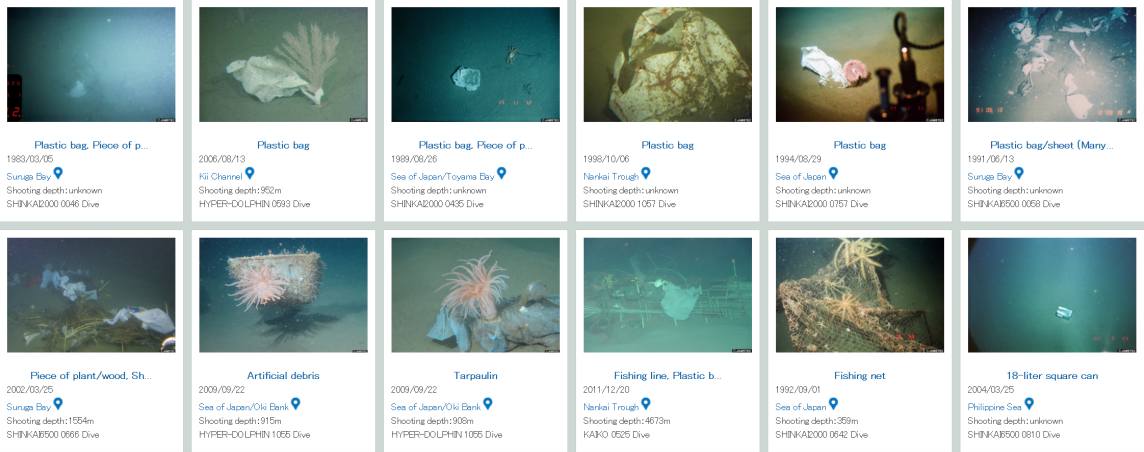


Figure 14: Sample of dataset

Dataset:

J-EDI dataset with 7,212 images containing annotated bounding boxes and segmentation masks for different types of marine debris. Two classification versions: Material-based and Instance-based. Dataset Link

Task: Implement a deep learning model for automated marine debris detection. Utilize pre-trained object detection models like Faster R-CNN and Mask R-CNN to analyze underwater images. Train and evaluate models using benchmark datasets, optimize detection accuracy, and assess the effectiveness of different architectures for real-world deployment.

Relevant Papers:

[1]. Hong, J., Fulton, M., & Sattar, J. (2020). TRASHCAN: A Semantically-Segmented Dataset for Marine Debris Detection. *arXiv preprint arXiv:2007.08097*. Paper Link

[2]. Valdenegro-Toro, M. (2016). Submerged Marine Debris Detection with Autonomous Underwater Vehicles. *RAHA 2016*. Paper Link

[3]. Ge, Z., Shi, H., Mei, X., Dai, Z., & Li, D. (2016). Semi-Automatic Recognition of Marine Debris on Beaches. *Scientific Reports*. Paper Link

015. Cell Tracking Challenge

Area of Research: Computer Vision, Object Checking, Biomedical image analysis

Problem Statement: Cell tracking is a critical task in biomedical research and medicine, facilitating the study of cellular dynamics, understanding disease progression, and evaluating therapeutic interventions. Accurate tracking of cells over time in sequence imaging data is necessary to elucidate complex biological processes and interactions.

The primary challenge in cell tracking arises from various technical and biological factors. These include high cell density in images, the presence of noise, variations in cell shape and size, cell division, and movement, as well as the heterogeneity in cell appearance due to different imaging modalities. Current methodologies often struggle with robustness and accuracy, particularly in real-time applications and in scenarios involving large datasets. Improving cell tracking techniques can significantly enhance our understanding of cellular mechanisms underlying health and disease. For instance, better tracking algorithms can lead to more accurate studies of cancer cell metastasis, immune cell interactions, and tissue regeneration processes. Additionally, advancements in this area could improve the efficacy of personalized medicine by enabling precise monitoring of cellular responses to treatments.

In this challenge, you need to develop segmentation and tracking methods using either real or computer-generated (2D or 3D) time-lapse microscopy videos of cells and nuclei. This is an intermediate-level course project that requires an understanding of instance segmentation networks and object tracking networks.

Dataset:

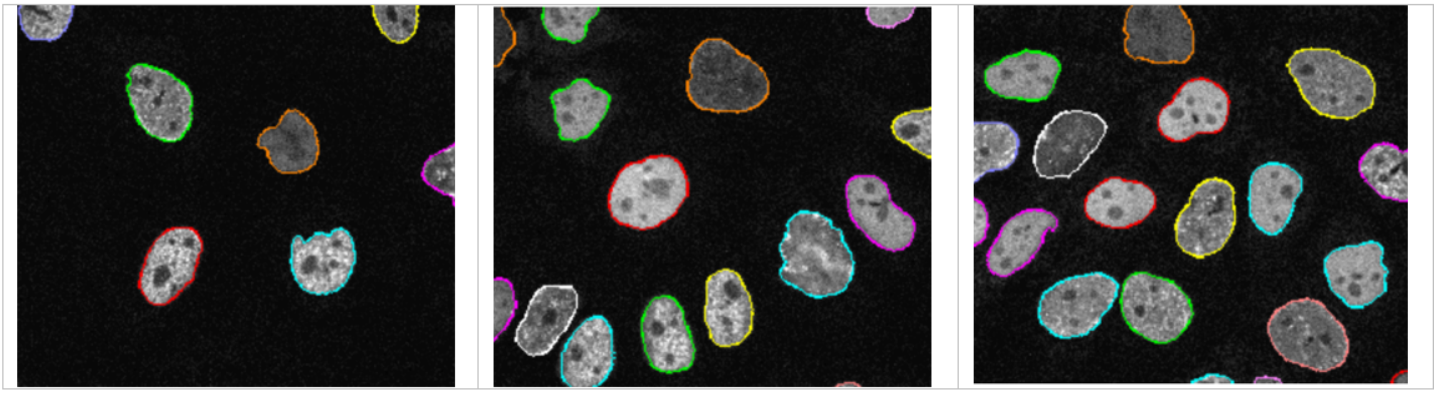


Figure 15: Examples from the dataset for cell tracking problem.

Dataset URL: https://celltrackingchallenge.net/2d-datasets/ and https://celltrackingchallenge.net/3d-datasets/

Task: Build a Deep Learning model for Cell Tracking Problem.

Relevant Papers

[1] Maška, M., Ulman, V., Delgado-Rodriguez, P., Gómez-de-Mariscal, E., Nečasova, T., Guerrero Pena, F.A., Ren, T.I., Meyerowitz, E.M., Scherr, T., Löffler, K. and Mikut, R., 2023. The cell tracking challenge: 10 years of objective benchmarking. *Nature Methods*, *20*(7), pp.1010-1020.

[2] Ulman, V., Maška, M., Magnusson, K.E., Ronneberger, O., Haubold, C., Harder, N., Matula, P., Matula, P., Svoboda, D., Radojevic, M. and Smal, I., 2017. An objective comparison of cell-tracking algorithms. *Nature methods*, *14*(12), pp.1141-1152.

[3] Meijering, E., 2020. A bird's-eye view of deep learning in bioimage analysis. *Computational and structural biotechnology journal*, *18*, pp.2312-2325.

[4] Urdiales, J., Martin, D. and Armingol, J.M., 2023. An improved deep learning architecture for multi-object tracking systems. *Integrated Computer-Aided Engineering*, *30*(2), pp.121-134.